

Active, phosphorylation-dependent mitogen-activated protein kinase (MAPK/ERK), stress-activated protein kinase/c-Jun N-terminal kinase (SAPK/JNK), and p38 kinase expression in Parkinson's disease and Dementia with Lewy bodies

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Summary. The expression of mitogen-activated protein kinases, extracellular signal-regulated kinases (MAPK/ERK), stress-activated protein kinases, c-Jun N-terminal kinases (SAPK/JNK), and p38 kinases is examined in Parkinson disease (PD), in Dementia with Lewy bodies (DLB), covering common and pure forms, and in age-matched controls. The study is geared to gaining understanding about the involvement of these kinases in the pathogenesis of Lewy bodies (LBs) and associated *tau* deposits in Alzheimer changes in the common form of DLB. Active, phosphorylation dependent MAPK (MAPK-P) is found as granular cytoplasmic inclusions in a subset of cortical neurons bearing abnormal *tau* deposits in common forms of DLB. Phosphorylated p-38 (p-38-P) decorates neurons with neurofibrillary tangles and dystrophic neurites of senile plaques in common forms of DLB. Phosphorylated SAPK/JNK (SAPK/JNK-P) expression occurs in cortical neurons with neurofibrillary tangles in the common form of DLB. Lewy bodies (LBs) in the brain stem of PD and DLB are stained with anti-ERK-2 antibodies, but they are not recognized by MAPK-P, SAPK/JNK-P and p-38-P. Yet MAPK-P, p-38-P and SAPK/JNK-P immunoreactivity is found in cytoplasmic granules in the vicinity of LBs or in association with irregular-shaped or diffuse α -synuclein deposits in a small percentage of neurons, not containing phosphorylated *tau*, of the brain stem in PD and DLB. MAPK-P, p-38-P and SAPK-P are not expressed in cortical LBs or in cortical neurons with α -

synuclein-only inclusions in DLB. MAPK-P, p-38-P and SAPK/JNK-P are not expressed in α -synuclein-positive neurites (Lewy neurites) in PD and DLB as revealed by double-labeling immunohistochemistry. These results show that MAPKs are differentially regulated in neurons with α -synuclein-related inclusions and in neurons with abnormal *tau* deposits in DLB. Moreover, different kinase expression in brain stem and cortical LBs suggest a pathogenesis of brain stem and cortical LBs in LB diseases. Finally, no relationship has been observed between MAPK-P, p-38-P and SAPK/JNK-P expression and increased nuclear DNA vulnerability, as revealed with the method of in situ end-labeling of nuclear DNA fragmentation, and active, cleaved caspase-3 expression in neurons and glial cells in the substantia nigra in PD and DLB.

Keywords: α -synuclein, Dementia with Lewy bodies, ERK, kinase, MAPK, Parkinson, p-38, SAPK, *tau*.

Introduction

Parkinson's disease (PD) and Dementia with Lewy bodies (DLB) are characterized by the presence of intracytoplasmic inclusions called Lewy bodies (LBs) localized in neurons of the substantia nigra pars compacta, locus ceruleus and other selected nuclei of the brain stem, as well as several nuclei of the basal forebrain and hippocampus. PD and DLB are considered LB diseases. LBs of cortical type are extended to neurons of the entorhinal cortex, cerebral isocortex and diencephalon in DLB (Kosaka, 1993; Pollanen et al., 1993; Forno, 1996; Lennox and Lowe, 1998; Ince et al., 1998; Galvin et al., 1999). Ultrastructurally, brainstem-type LBs are composed of a dense core surrounded by radiating thick and thin fibrils. Cortical-type LBs are composed of loosely arranged fibrils with no distinct dense cores (Duffy and Tennyson, 1965; Pappolla, 1986; Kuzuhara et al., 1998; Forno et al., 1986). LBs contain phosphorylated and non-phosphorylated neurofilaments (Baner et al., 1989; Hill et al., 1991; Schmidt et al., 1991; Galloway et al., 1992) and ubiquitin (Kuzuhara et al., 1988; Baner et al., 1989; Galloway et al., 1992; Manetto et al., 1988; Lennox et al., 1989). α -synuclein, a cytosolic protein abundant in presynaptic terminals, is another major component of brainstem and cortical LBs (Spillantini et al., 1997; Wakabayashi et al., 1997; Baba et al., 1998; Spillantini et al., 1998; Irizarry et al., 1998; Takeda et al., 1998; Arima et al., 1998; Trojanowski et al., 1998; Hashimoto and Masliah, 1999). Alzheimer-type pathology, including senile plaques and neurofibrillary degeneration with abnormal *tau* deposition, is a common associated feature in DLB (Ince et al., 1998; Gómez-Tortosa et al., 2000). LBD with no AD changes is known as the pure form of DLB, whereas LBD with associated AD changes is recognized as the common form of DLB (Kosaka, 1993).

α -synuclein is constitutively phosphorylated at Ser129 and Ser87 (Okochi et al., 2000). Casein-kinases 1 and 2 (CK-1, CK-2) have the capacity to phosphorylate Ser129 (29). Fyn phosphorylates α -synuclein at Tyr125 (Nakamura et al., 2001). Therefore, CK-1, CK-2 and Fyn may also regulate α -synuclein phosphorylation in LBs. Studies in vitro have also shown synucleins as putative substrates for G protein-coupled receptor kinases (Pronin et al., 2000). In

addition, other kinases, including cyclin-dependent kinase 5 (Cdk5) and calcium/calmodulin-dependent kinase-II (CaM Kinase-II), probably participate in the phosphorylation of various components of LBs, as Cdk5 and CaM Kinase-II are expressed in both cortical- and brainstem-type LBs bodies (Iwatsubo et al., 1991; Okochi et al., 2000; Brion and Couck, 1995; Takahashi et al., 2000). It is not known whether other kinases might be involved in LB formation.

The family of mitogen-activated protein kinases (MAPKs) is composed of different members, including extracellular signal-regulated kinases (ERKs), and stress-activated protein kinases (SAPKs), c-Jun N-terminal kinases (JNKs) and p38 kinases (Cobb and Goldsmith, 1995; Robinson and Cobb, 1997; Mielke and Herdegen, 2000; Ip and Davis, 1998; Nebreda and Porras, 2000). Previous studies have shown the involvement of active, phosphorylation-dependent MAPK/ERK in *tau* phosphorylation in neurons in Alzheimer disease (AD), and in neurons and glial cells in Pick's disease, progressive supranuclear palsy and corticobasal degeneration (Perry et al., 1999; Ferrer et al., 2001). Studies in AD have demonstrated strong non-phosphorylated-p38 and phosphorylated-p38 immunoreactivity in neurons with neurofibrillary degeneration and dystrophic neurites surrounding senile plaques (Zhu et al., 2000; Ferrer et al., 2002). Moreover, active, phosphorylation-dependent p38 immunoreactivity is found in the majority of *tau* deposits in neurons and glial cells in tauopathies (Ferrer et al., 2002). Finally, strong phosphorylated SAPK/JNK immunoreactivity has been reported in association with phosphorylated *tau* deposits in neurons and glial cells in AD and other tauopathies (Ferrer et al., 2002).

The present study examines the expression of ERK, and phosphorylated MAPK/ERK, p38 and SAPK/JNK in the brains of PD, and pure and common forms of LBD, in order to gain understanding about the possible involvement of these kinases in LBs and related α -synuclein deposits in PD and DLB in comparison with neurofibrillary pathology and related phosphorylated *tau* deposits in the common form of LBD.

Material and methods

Human cases

Samples of five patients with PD, nine patients with DLB and five age-matched controls were obtained at post-mortem. The mean duration of PD was 14 years. The mean duration of DLB was 8.6 years. The causes of death were similar in control and neurodegenerative cases. Patients with long agonal states were not included in this series. The post-mortem delay was between 2h and 7h in control and diseased brains.

The complete neuropathological examination was carried out in every case (diseased cases and controls) in formalin-fixed tissue for no less than three weeks; the tissue was then embedded in paraffin. The following regions were analyzed: middle frontal gyrus at the plane just rostral to the temporal tip, middle temporal gyrus at the plane of the mammillary body, intraparietal sulcus at the plane 1 cm caudal to the posterior pole of the splenium of the corpus callosum, anterior cingulate gyrus at the level of the anterior commissure, parahippocampal gyrus at the plane of the red nucleus, hippocampus and dentate gyrus, dorsal thalamus, amygdala, nucleus basalis of Meynert, caudate and puta-

men, pallidum, midbrain including the substantia nigra, pons including locus ceruleus, medulla oblongata including the dorsal vagus nerve.

Dewaxed sections, 7 µm thick, were stained with haematoxylin and eosin, luxol fast blue-Klüver Barrera and methenamine silver (PAM), or processed for immunohistochemistry following the avidin-biotin-peroxidase method (ABC kit, Vectastain, Vector). Antibodies to phosphorylated neurofilaments of 170 kDa or 200 kDa (clones BF10 and RT97, Boehringer-Mannheim) were used at dilutions of 1:100 and 1:50, respectively. Antibodies to *tau* (Sigma) were used at a dilution of 1:10. The *tau* antibody is derived from the hybridoma produced by fusion of mouse myeloma cells and splenocytes from an immunized mouse. Purified bovine microtubule associated proteins were used as immunogens. Several electrophoretic bands were encountered on SDS polyacrylamide gels (55 kDa–62 kDa). The *tau* antibody recognizes phosphorylated *tau* in paraffin sections. Rabbit polyclonal antibodies to α -synuclein (Chemicon) were used at a dilution of 1:1000. Antibodies to glial fibrillary acidic protein (GFAP, Dako, Dakopats), β A4-amyloid (Boehringer-Mannheim) and ubiquitin (Dako) were used at dilutions of 1:250, 1:5 and 1:200, respectively. Sections immunostained with antibodies to α -synuclein and β A4-amyloid were enhanced by pre-incubation of the sections with 30% formic acid for 5 min.

Diagnostic of LBD according to Lewy body scoring was carried out following the consensus guidelines of the Consortium of DLB international workshop (McKeith et al., 1996). Staging of AD changes was performed following Braak and Braak criteria (1999). DLB cases were categorized as common (six cases) or pure (three cases) forms on the basis of the presence or absence, respectively, of associated AD pathology (Kosaka, 1993; Lennox and Lowe, 1998; Ince et al., 1998; McKeith et al., 1996). A summary of the cases with the main neuropathological findings is shown in Table 1.

Table 1. Summary of the main clinical and pathological findings. α -synuclein pathology (presence of Lewy bodies: LBs) is categorized following the consensus guidelines of the consortium on DLB International Workshop (McKeith et al., 1996). *tau* pathology follows the stages of Alzheimer disease changes of Braak and Braak (see Braak and Braak, 1999). Diagnosis is based on the clinical manifestations and neuropathological data as PD: Parkinson disease, and DLB pure or common: pure and common forms of Dementia with Lewy Bodies (Kosaka, 1993; McKeith et al., 1996; Ince et al., 1998)

Case	Age	Gender	α -synuclein: LBs	β A4 amyloid (plaques)	<i>tau</i> pathology	Diagnosis
1	78	F	Brain stem	+	0	PD
2	73	F	Brain stem	+	0	PD
3	71	F	Brain stem	0	0	PD
4	68	M	Brain stem	0	0	PD
5	69	M	Brain stem	0	0	PD
6	68	F	Neocortical	0	0	DLB pure
7	66	F	Neocortical	+	I	DLB pure
8	74	F	Neocortical	+++	III–IV	DLB common
9	82	F	Neocortical	+++	IV	DLB common
10	78	F	Neocortical	+++	V–VI	DLB common
11	65	M	Neocortical	+	0	DLB pure
12	75	M	Neocortical	+++	V	DLB common
13	78	M	Neocortical	++	V	DLB common
14	78	M	Neocortical	++	IV–V	DLB common
15	68	F	0	0	0	Control
16	72	F	0	++	II–III	Control
17	72	F	0	+	0	Control
18	70	M	0	0	0	Control
19	76	M	0	+	I	Control

Kinase immunohistochemistry

For specific immunohistochemical studies, fresh samples of the hippocampus and entorhinal cortex, anterior cingulate gyrus, frontal cortex, midbrain including substantia nigra, and pons including locus ceruleus were fixed with 4% paraformaldehyde in phosphate-buffered saline (PBS) for 48h, washed in PBS and embedded in paraffin. Immunohistochemistry was carried out following the ABC method. Dewaxed sections were first boiled in citrate buffer and then stored overnight at room temperature. After blocking endogenous peroxidase with methanol and hydrogen peroxide, the sections were incubated with normal serum to block unspecific binding and then incubated at 4°C overnight with one of the primary antibodies.

The anti-ERK mouse monoclonal antibody (E17120, Transduction Laboratories) is raised against amino acids 219–358 at the carboxy terminal region of rat ERK-2. The antibody recognizes a unique band of 42kDa on Western blots of human samples that corresponds to the molecular weight of ERK-2. The anti-MAP kinase phospho-specific (MAPK-P) rabbit polyclonal antibody (Calbiochem) is raised against a synthetic phospho-tyrosine peptide corresponding to residues 196–209 of human p44 MAP kinase. The antibody detects phosphorylated Tyr204 of p44 and p42 MAP kinases (phosphorylated ERK-1 and ERK-2). The purified rabbit polyclonal phospho-SAPK/JNK (Thr183/Tyr185) antibody (SAPK/JNK-P) (Cell Signaling) is produced against a synthetic phospho-Thr183/Tyr185 peptide corresponding to the residues of human SAPK/JNK. The purified phospho-p38 MAP kinase (Thr180/Tyr182) (p38-P) rabbit polyclonal antibody (Cell Signaling) detects p38 MAP kinase only when activated by dual phosphorylation at Thr180 and Tyr182. The antibodies were used at dilutions of 1:2000, 1:100, 1:100 and 1:200, respectively for ERK-2, MAPK-P, SAPK/JNK-P and p-38-P.

Following incubation with the primary antibody, the sections were incubated with biotinylated anti-rabbit or anti-mouse IgG, diluted 1:100, and then with ABC at a dilution of 1:100 for 1h each step at room temperature. The peroxidase reaction was visualized with 0.05% diaminobenzidine and 0.01% hydrogen peroxide. Tissue from control and diseased brains was processed in parallel to avoid day-to-day variations in the staining procedure. Sections were counterstained with haematoxylin.

Double-labeling immunohistochemistry was conducted by incubating the sections following a two-step protocol. The sections were first incubated with antibodies to one of the kinases, and the immunoreaction was visualized with diaminobenzidine and hydrogen peroxide as before. Subsequently, the sections were incubated with anti-*tau* (Sigma) at a dilution of 1:10, or anti- α -synuclein (Chemicon) at a dilution of 1:500, or anti-phosphorylated neurofilaments of 170kDa or 200kDa (Boehringer-Mannheim) at dilutions of 1:100 and 1:50, respectively. The immunoreaction was visualized with 0.01% benzidine hydrochloride, 0.025% sodium nitroferricyanide in 0.0001% M sodium phosphate buffer (pH 6.0), and 0.005% hydrogen peroxide, or with NH_4NiSO_4 (0.05M) in phosphate buffer (0.1M), diaminobenzidine, NH_4Cl and hydrogen peroxide. The first primary antibody was recognized as a brown homogeneous precipitate, whereas the second primary antibody was recognized as a dark blue (almost black when using NH_4NiSO_4) precipitate.

In situ end-labeling of nuclear DNA fragmentation and caspase-3 immunohistochemistry

Some sections of the brain stem were stained with the method of in situ end-labeling of nuclear DNA fragmentation with ApopTag: in situ apoptosis detection kit (Oncor) following the instructions of the supplier. In addition, active caspase-3 immunohistochemistry was carried out by using a cleaved caspase-3 (17kDa) rabbit polyclonal antibody (Cell Signaling) produced with a synthetic peptide (KLH coupled) corresponding to residues surrounding the cleavage site of human caspase-3. The antibody, used at

a dilution of 1:100 following the ABC method, detects only the large fragment of activated caspase-3. It does not recognize full-length caspase-3 or other caspases. Cleaved, active caspase-3 expression has been used here as a marker of caspase-3-dependent apoptotic cell death (Nicholson and Thornberry, 1997; Desagher and Martinou, 2000).

Double-labeling with the method of in situ end-labeling of nuclear DNA fragmentation or cleaved caspase-3 and ERK-2, MAPK-P, SAPK/JNK-P or p38-P immunohistochemistry was carried out following a two-step protocol, as previously.

Results

ERK-2, MAPK-P, p-38-P and SAPK/JNK-P immunohistochemistry in AD-associated pathology in DLB

Weak ERK-2 immunoreactivity was seen in the cytoplasm of neurons of the hippocampus and cerebral cortex in control and diseased brains. No differences in ERK-2 expression were seen in neurons with neurofibrillary tangles and in dystrophic neurites of DLB cases.

MAPK-P expression in control aged-matched brains was restricted to weak diffuse immunoreactivity in the soma of scattered neurons. A number of cortical neurons, including neocortex, cingulate cortex, entorhinal cortex and hippocampus, in DLB with associated AD pathology were stained with MAPK-P antibodies (Fig. 1A,B). MAPK-P immunoreactivity occurred as cytoplasmic granules only in neurons that contained abnormal *tau* deposition, as revealed by double-labeling immunohistochemistry (Fig. 1C). Not all neurons containing *tau* deposits were stained with anti-MAPK-P antibodies. In fact, only about 20% of neurons with neurofibrillary tangles contained MAPK-P-positive cytoplasmic granules. Neurofibrillary tangles were not stained with anti-MAPK-P antibodies. Dystrophic neurites of senile plaques were not stained with the anti-MAPK-P antibody.

p-38-P expression in control aged-matched brains was restricted to weak diffuse immunoreactivity in the soma of scattered neurons. Large numbers of cortical neurons showed p38-P-immunoreactive cytoplasmic granular inclusions in DLB with associated AD pathology (Fig. 1D). In contrast to MAPK-P, the anti-p38-P antibody decorated neurofibrillary tangles (Fig. 1E). Dystrophic neurites of senile plaques were also stained with anti-p38-P antibodies (data not shown). Double-labeling immunohistochemistry showed that 80% of p-38-P-immunoreactive cortical neurons co-localized phosphorylated *tau* and, inversely, almost all *tau*-positive neurons co-localized p-38-P (Fig. 1F).

SAPK/JNK-P immunoreactivity was restricted to a small number of nuclei of cortical and subcortical neurons in control and diseased brains. In addition, SAPK/JNK-immunoreactivity decorated neurofibrillary tangles in DLB cases with associated AD pathology (Fig. 1G,H). Positive cells co-localized phosphorylated *tau* as revealed with double-labeling immunohistochemistry (Fig. 1I). Yet not all neurons with *tau* deposits were stained with anti-SAPK/JNK-P antibodies; about 50% of the neurofibrillary tangles were decorated with anti-SAPK/JNK-P antibodies.

*ERK-2, MAPK-P, p-38-P and SAPK/JNK-P immunohistochemistry in
LB-associated pathology in PD and DLB*

LBs in PD and DLB were easily visualized with anti- α -synuclein antibodies (Fig. 2A). In addition to LBs, moderate to strong α -synuclein immunoreactivity was observed in irregular-shaped inclusions and, diffusely, in the cytoplasm of scattered neurons (Fig. 2B), as well as in abnormal neurites (Lewy neurites). Brain stem LBs were also stained with antibodies to phosphorylated NFs of high molecular weight (data not shown).

Moderate ERK-2 immunoreactivity was seen in LBs of the substantia nigra and other nuclei of the brain stem in PD and LBD (Fig. 2C). Irregular-shaped hyaline inclusions were also decorated with anti-ERK-2 antibodies (Fig. 2D). LBs were not stained with antibodies to MAPK-P, p-38-P or SAPK/JNK-P in LB and DLB. However, punctate or granular MAPK-P immunoreactivity occurred at the periphery of a minority of LBs or in the cytoplasm of nigral neurons containing LBs (Fig. 2E).

Granular p-38-P immunoreactivity was found in neurons of the substantia nigra with LBs (Fig. 2F), but also in neurons with no apparent LB formation in sections counterstained with haematoxylin (Fig. 2G). Neurons with LBs were rarely stained with anti-SAPK/JNK-P antibodies (Fig. 2H). Interestingly, double-labeling immunohistochemistry disclosed p-38-P immunoreactivity in association with diffuse α -synuclein deposition in selected neurons of the substantia nigra (Fig. 2I).

In no case, MAPK-P, p-38-P and SAPK/JNK-P immunoreactivity in nigral neurons was associated with abnormal *tau* deposition as revealed with double-labeling immunohistochemistry.

Because of the few numbers of neurons showing phosphorylated kinase expression, consecutive alternate sections of the brain stem processed for single and double-labeling immunohistochemistry were examined in every case. Quantitative studies disclosed that only about 10% and 20% of neurons of the substantia nigra with α -synuclein pathology were stained with anti-MAPK-P and anti-p38-P antibodies, respectively.

In contrast with brain stem neurons with LBs and related inclusions, cortical neurons with LBs were not decorated with anti-ERK-2, MAPK-P, SAPK/JNK-P and p-38-P antibodies despite careful search in consecutive sections processed for single and double-labeling immunohistochemistry.

Lewy body neurites, distinguished by their positive α -synuclein staining, were not decorated with anti ERK-2, MAPK-P, p-38-P and SAPK/JNK-P antibodies.

*Caspase-3 immunohistochemistry and double-labeling of nuclear DNA
fragmentation and kinase immunohistochemistry*

The method of in situ end-labeling of nuclear DNA fragmentation disclosed only a few positive cells consistent with glia in the vicinity but not within the substantia nigra in control and diseased brains. Apoptotic figures were not seen in any case. In agreement with the lack of cells with apoptotic morphology, cleaved caspase-3 immunohistochemistry barely disclosed a single

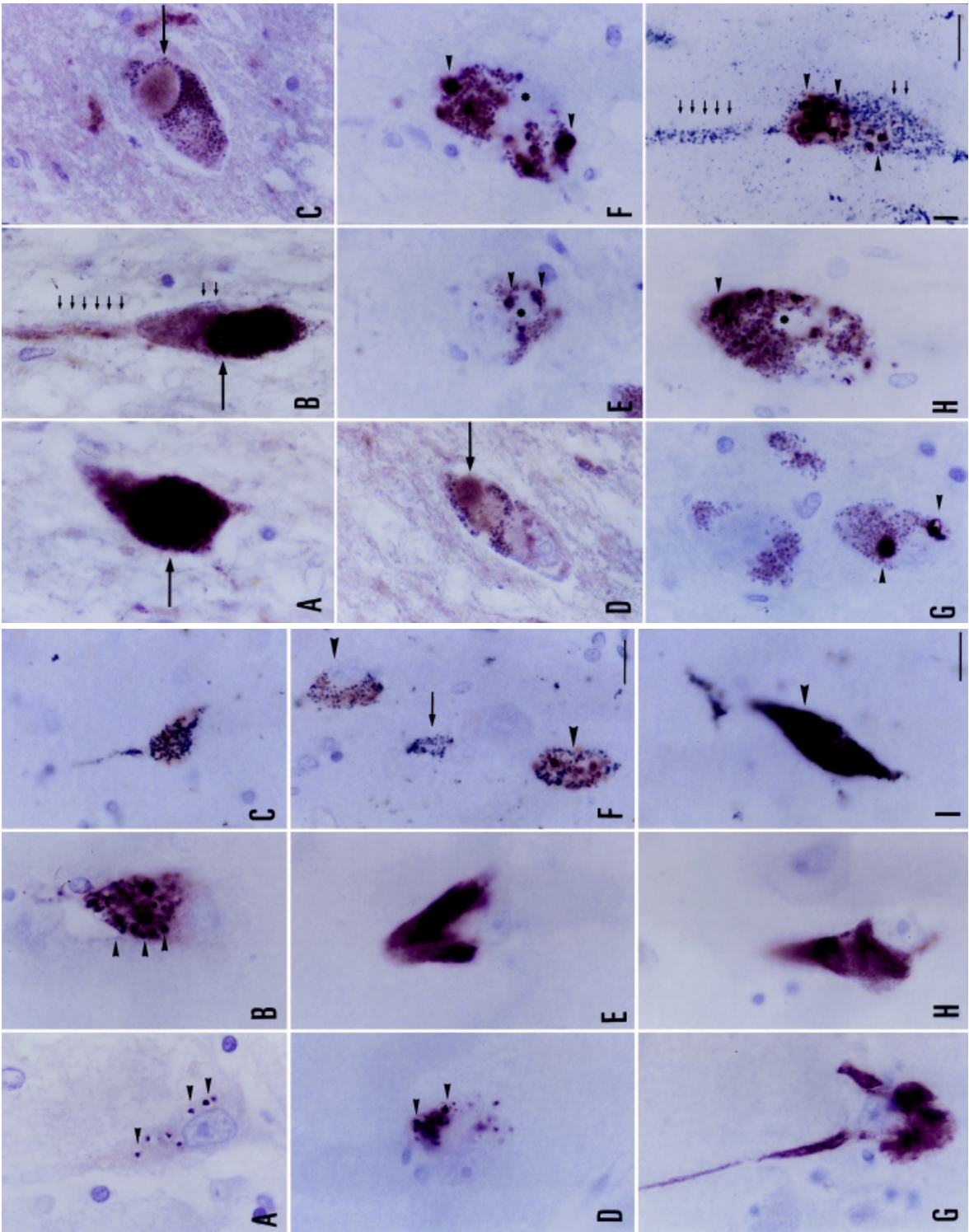


Fig. 1

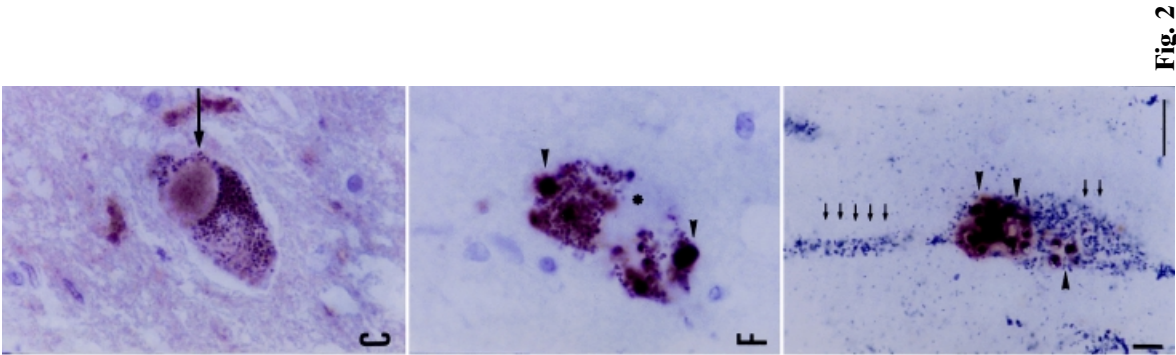


Fig. 2

positive cell in sections containing the substantia nigra, but no a single positive cell was seen in the substantia nigra itself. However, the method was sensitive, as many active caspase-3-immunoreactive cells were seen in tissue sections from human medulloblastomas that were processed in parallel, in which apoptosis was commonly observed (data not shown).

Double-labeling of nuclear DNA fragmentation and kinase immunohistochemistry disclosed no relationship between cells bearing nuclear DNA breaks and cells with ERK-2, MAPK-P, SAPK/JNK-P and p38-P immunoreactivity (data not shown).

Discussion

Studies in vitro have shown that MAPK/ERKs phosphorylate *tau* and neurofilament proteins of medium and high molecular weight at Lys-Ser-Pro (KSP) motifs (Roder and Ingram, 1991; Drewes et al., 1992; Goedert et al., 1992; Ledesma et al., 1992; Veeranna et al., 1998; Goedert et al., 1998). SAPKs are implicated in the phosphorylation of Lys-Ser-Pro motifs in neurofilaments and *tau* (Giasson and Mushynski, 1996; Goedert et al., 1997; Reynolds et al., 1997a). p38 also has the capacity to phosphorylate *tau* protein in vitro (Reynolds et al., 1997a, 2000).

Previous studies have shown strong ERK-2 and MAPK-P immunoreactivity in a small percentage of neurons containing *tau* deposits in AD (Trojanowski et al., 1993; Perry et al., 1999; Ferrer et al., 2001). In contrast, strong p38-P and SAPK/JNK-P immunoreactivity occurs in a large number of neurons with neurofibrillary tangles and in dystrophic neurites of senile

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Fig. 1. MAPK-P (A–C), p-38-P (D–F) and SAPK/JNK-P (G–I) immunoreactivity in cortical neurons in DLB with associated AD pathology. MAPK-P-immunoreactive cytoplasmic granules are seen in scattered neurons of the hippocampus (A) and frontal cortex (B) that co-localize abnormal *tau* deposition as revealed with double-labeling immunohistochemistry (C). p-38-P expression is found in cytoplasmic granules (D), but also in neurofibrillary tangles (E), in the majority but not all, neurons containing abnormal *tau* deposition as disclosed in double-immunostained sections (F). SAPK/JNK-P antibodies decorate neurofibrillary tangles (G, H) which are immunoreactive to anti-*tau* (I). Kinases are recognized as brown precipitates, whereas *tau* deposition is seen as a dark blue precipitate. Paraffin sections, haematoxylin counterstaining. A, B, D, E, G, H, I, bar in I = 10 µm; C, F, bar in F = 25 µm

Fig. 2. Representative fields of the substantia nigra in PD (A, C, E, F, H) and DLB (B, D, G, I). LBs are recognized by anti- α -synuclein antibodies (A). In addition, abnormal α -synuclein deposition takes the form of irregular-shaped inclusions and diffuse deposits in the cytoplasm and proximal dendrites (small arrows) of brain stem neurons (B). Brain stem LBs and irregular-shaped inclusions are stained with anti-ERK-2 antibodies (C, D). MAPK-P (E), p-38-P (F, G) and SAPK/JNK-P (H) immunoreactivity (arrowheads) is seen in the vicinity of LBs or irregular-shaped inclusions (asterisks), and in neurons apparently lacking LBs. Yet, double-labeling immunohistochemistry discloses p-38-P immunoreactivity (brown precipitate) in association with abnormal α -synuclein deposition (dark blue precipitate) in the cytoplasm and dendrites (small arrows) (I). Paraffin sections; haematoxylin counterstaining; bar = 10 µm

plaques in AD (Zhu et al., 2000; Ferrer et al., 2002). MAPK-P, SAPK/JNK-P and p-38-P is also expressed in neurons and glial cells containing abnormal *tau* deposits in other tauopathies as Pick's disease, progressive supranuclear palsy and corticobasal degeneration (Ferrer et al., 2001, 2002). These observations suggest that MAPK-P, p38-P and SAP/JNK-P may play a role throughout the process of *tau* phosphorylation in AD and other tauopathies. The present findings further support a function for MAPK/ERK, p38 and SAPK/JNK in *tau* phosphorylation in AD-associated changes in the common forms of DLB. Furthermore, the presence of p-38-P and SAPK/JNK-P in neurons with neurofibrillary degeneration suggests sequestration of phosphorylated kinase isoforms, which are usually short-lived proteins, in association with abnormal *tau* deposits.

In contrast with the extended kinase expression in association with abnormal *tau* deposits in AD-related changes in DLB, the present results have shown ERK-2, MAPK-P, SAPK/JNK-P and p38-P immunoreactivity in a limited number of neurons of the brain stem bearing LBs or diffuse deposits of α -synuclein and phosphorylated neurofilaments. Yet these findings may suggest a role for MAPK/ERK, SAPK/JNK and p-38 in the phosphorylation of undetermined components of LBs in PD and DLB. It has been shown that α -synuclein links to ERK-2, and that this may have effects on Elk-1 phosphorylation (Iwata et al., 2001). However, MAPK and Cdk5, in contrast to Fyn, do not phosphorylate α -synuclein in vitro (Nakamura et al., 2001). Therefore, it is likely that MAPK, SAP/JNK and p-38 participate at appropriate stages of phosphorylation of recruited proteins in diffuse deposits associated with α -synuclein or at the periphery of LBs in brain stem neurons. This is further supported by the co-localization of specific kinases and phosphorylated neurofilament deposits in selected neurons. Interestingly, Cdk5 is also involved in the phosphorylation of neurofilaments and microtubules in vitro (Hisanaga et al., 1995; Pant and Veeranna, 1995). Cdk5 has also been reported as present in LBs in PD and LBD, thus suggesting Cdk5 involvement in neurofilament phosphorylation of LBs (Brion and Couck, 1995; Takahashi et al., 2000). Moreover, Cdk5 and MAPKs are associated with complexes of cytoskeletal proteins in rat brain (Veeranna et al., 2000). Taken together, these observations suggest a complex interaction of kinases in the phosphorylation of protein components linked with brain stem LBs. Yet the identification of putative phosphorylation substrates of MAPKs in PD and DLB needs to be elucidated.

The morphology and composition of LBs is subject to regional differences. Brain stem LBs differ from cortical LBs (Duffy and Tennyson, 1965; Pappolla, 1986; Kuzuhara et al., 1988; Forno et al., 1986; Pollanen et al., 1993), although ubiquitin and α -synuclein are common components of brain stem and cortical LBs. Active, phosphorylation-dependent MAPKs are expressed in target brain stem neurons in PD and DLB, but not in cortical LB in DLB. These data suggest a differential pathogenesis of brain stem and cortical LBs in LB diseases.

Finally, MAPK, SAPK/JNK and p-38 have been implicated in apoptosis underlying several neurodegenerative diseases (Oh-hashii et al., 1999; Bhat

and Zhang, 1999; Mielke and Herdegen, 2000). Previous studies have shown no relationship between MAPK/ERK-P, SAPK/JNK-P and p-38-P expression and increased nuclear DNA vulnerability in *tau*-positive inclusions in AD and other tauopathies (Ferrer et al., 2001, 2002). In line with these observations, the present results have shown no relationship between phosphorylated kinase expression and DNA vulnerability, as revealed with the method of in situ end-labeling of nuclear fragmentation, and active caspase-3 expression in the substantia nigra in PD and DLB. Moreover, the present findings further support previous observations showing the presence of scattered glial cells but not a single neuron with nuclear DNA breaks or with cleaved caspase-3 expression in the substantia nigra in PD and DLB (Jellinger, 2000). These findings make unlikely apoptosis as the cause of neuron death in LBDs.

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